The Physics Substrate — Cavities, Materials, Oscillators and Gating

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Layer 1. How a concept becomes a mechanical object and a relation becomes a material.

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PRM: The Phonic Resonance Matrix — Concepts as Resonant Cavities and Inference as Wave Propagation

"Inhibition is not an `if` statement. It is two waves arriving out of phase."

Abstract

We specify a substrate in which a concept is not an identifier, a token or an embedding, but a **physical resonant cavity** with mass, displacement and velocity. Relations are not pointers but **tunnels** that transmit mechanical force between cavities. Inference is not search but **wave propagation**, and the answer to a query is whichever cavity is vibrating hardest once the field has settled.

The architecture's distinguishing property is that several capabilities usually requiring explicit algorithms become consequences of the medium. Inhibition is destructive interference rather than a conditional. Multi-constraint search is two wavefronts colliding rather than a set intersection. Structural analogy is sympathetic resonance between cavities that share a harmonic ratio and no edge at all.

Section 7 reports what was measured and what remains an argument from mechanism, including one claim in the original formulation that does not survive scrutiny.

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1. The Substrate

Concepts are **resonant volumes**. Each cavity carries a mass, a displacement x, a velocity v, and a natural frequency determined by its physical parameters. In the acoustic reading, a large cavity has a low resonant frequency and a small one a high frequency.

Relations are **physical tunnels** — channels connecting cavities, not references. Two polarities exist:

- An **excitatory** tunnel transmits displacement directly, producing sympathetic resonance in the target cavity.
- An **inhibitory** tunnel inverts phase by 180°, so its contribution actively cancels vibration at the target.

The design commitment is that the state of a concept is its *actual mechanical state*, not a value standing for one.

---

2. Dynamics

The tick integrates Hooke's law and Newton's second law across every tunnel:

$$F \;=\; k\,(x_a \cdot p - x_b) \;-\; c\,(v_a - v_b)$$

$$v \;\mathrel{+}=\; \frac{F}{m}\,\Delta t, \qquad x \;\mathrel{+}=\; v\,\Delta t$$

where k is tunnel stiffness, c damping, $p \in \{+1, -1\}$ polarity, and forces are applied equal and opposite to the two endpoints.

A query is an injected driving force rather than a string. Reading the result means observing which cavities have the largest displacement after the field settles — the physical analogue of placing pressure sensors in the pipes and seeing which are shaking.

---

3. Why Inhibition Requires No Logic

In a conventional implementation, negation is a branch: *if the relation is inhibitory, subtract instead of add.* Here the polarity term p sits inside the force expression, so an inhibitory tunnel produces a 180°-shifted contribution by arithmetic alone.

The consequence is architectural rather than aesthetic. Because inhibition is multiplication by -1 rather than a conditional, the update path contains no branch on relation semantics — which is what permits the whole tick to be expressed as scalar arithmetic and saturating nonlinearities, with no control flow that depends on what a relation *means*.

---

4. Emergent Capabilities

These are the paper's substantive claims: behaviours obtained from propagation rather than from an algorithm written to produce them.

4.1 Constructive interference as multi-constraint search Drive two cavities simultaneously — say `type_2_diabetes` and `autophagy`.

A graph search attempts to trace a literal edge path between them and fails if none exists. The matrix instead radiates waves outward from both. The wavefronts split, reflect, and meet. At a distant cavity — `mtor` — the two arrive **in phase**, amplitudes add, and that cavity spikes.

The system did not know `mtor` was the bridge and did not search for it. It vibrated because it sits at the geometric intersection where two independent wavefronts collided. The conjunction was computed by the medium.

4.2 Sympathetic resonance as structural analogy Strike a 440 Hz tuning fork and a nearby string tuned to 880 Hz begins to sound, with no mechanical connection between them.

If cavity parameters reflect structural properties, then energy injected into one structure can excite another that shares a harmonic ratio **and has no connecting edge**. That is a structural analogy — a repurposing candidate — found through harmonics rather than traversal.

4.3 Standing waves as attractor discovery Run the tick for a few thousand iterations and kinetic energy settles into a standing wave: some cavities become nodes of near-zero motion through destructive interference, others antinodes of maximum vibration.

Standing wave patterns are **computationally irreducible**: one cannot predict where the pattern forms by inspecting the rules; the simulation must be run. When the field settles, it has resolved the equilibrium of a large number of competing constraints simultaneously, and the high-amplitude cavities are the result.

4.4 Detecting this without a classifier Novelty is observed rather than classified. After the field settles, scan for cavities with high amplitude that were **not** directly driven:

```
// After the wave has propagated, find the novelty
for (const auto& cavity : cavities) {
```

```
    if (cavity.displacement > THRESHOLD) {
        if (!was_injected(cavity.id)) {
            // A cavity is vibrating that we never excited directly:
            // an emergent resonance point.
            report(cavity.physical_label, cavity.displacement);
        }
    }
}
```

There is no `if/else` classifier deciding what counts as interesting. The condition is *energy arrived here and we did not put it here*, which is a statement about the field rather than about a category.

---

5. Position Against Pattern Matching

Property	Neural / graph search	PRM
Concept state	activation value or embedding	physical displacement and velocity
Inhibition	a rule or conditional	mechanical cancellation of opposed waves
Reachability	programmed paths or learned weights	radiated energy; waves collide unprogrammed
Multi-constraint	explicit set intersection	interference of two wavefronts
Analogy	similarity metric over representations	sympathetic resonance at harmonic ratio
Output	argmax over scores	whichever cavity is physically shaking

The claim is that energy in a physical medium **radiates** rather than following single logical lines, and that novelty arises from the unprogrammed collisions of those radiating waves.

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6. Implementation Note

The substrate is a one-dimensional wave simulator of the kind used in game audio and fluid engines — tension, mass and displacement integrated directly. No tensor library, no autodiff, and no linear-algebra framework is involved, because nothing in §2 is a matrix operation.

---

7. What Was Measured, and What Was Overstated

Implemented and load-bearing. Cavities with mass, displacement and velocity; polarity-carrying tunnels; the force integration of §2; and amplitude readout are the Layer-1 substrate of the current engine (`struct Cavity`, `physics.hpp`). The branchless property of §3 holds: the tick contains one guard branch and no conditional on relation type.

Measured. Attractor settling (§4.3) and the undriven-cavity novelty probe (§4.4) run and produce results. Multi-source interference (§4.1) occurs and drives the analogical bridging mechanism reported separately with 20-seed reproducibility data.

Overstated in the original formulation. Section 4.2's sympathetic resonance was presented as though cavity volume already encoded real structural properties. It does not. Cavity

parameters are derived from graph position and relation fan-out, **not** from molecular structure, so the harmonic ratio between two cavities carries no independent structural information about the entities they denote. The mechanism is sound; the specific claim that it discovers *chemical* analogies without edges is unsupported by anything implemented, and analogical bridging in the current engine works through shared-target intersection rather than harmonics.

Not implemented. The literal acoustic framing — transducers, piezoelectric sensors, physical pipes — is expository. The substrate is a numerical simulation, and no claim is made that a physical acoustic device was built.

---

8. Limitations

Settling is assumed. Section 4.3 assumes the field reaches a standing wave. A driven system may instead lock into a ground state and refuse to leave, which was later observed as a real failure mode requiring a dedicated escape mechanism.

Amplitude is not relevance. High displacement means energy arrived, which is not the same as being an answer. A highly-connected hub accumulates energy from every direction; this required a separate thermodynamic mechanism to suppress ubiquitous participants.

Cost scales with the field, not the query. Every tick integrates every tunnel, so a query costs the same whether it touches three concepts or three thousand. At large ontology sizes this dominates, and sparse iteration over only the active wavefront is the necessary and currently incomplete remedy.

---

9. Conclusion

PRM's contribution is the identification of a representation in which several operations normally requiring algorithms are instead consequences of propagation: inhibition as phase inversion, conjunction as interference, and constraint resolution as settling.

The parts that survived are those making no assumption about what the cavities *denote* — the dynamics, the polarity treatment, the readout. The part that did not survive assumed cavity parameters encoded external structure they were never given, which is a useful reminder that a physical metaphor stops being explanatory exactly where the physics stops being connected to the domain.

References

- Mahir, I. *MIMS: Mechanical Impedance and Material Stiffness.* — the material table parameterising these tunnels.
- Mahir, I. *HNCA: Hierarchical Neural Constraint Architecture.* — the macro-layer wrapping these cavities into frequency-separated regions.
- Mahir, I. *RB+G: Gated Recursive Bifurcation with Analogical Bridging.* — the mechanism that actually performs analogical discovery.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c.
- Implementation: `src/physics.hpp` (`struct Cavity`, `struct Tunnel`, `tick()`).

CMD: Continuous Morphological Dynamics — Sensory Grounding and Thermodynamic Habituation

> *"A stopword is not a word on a list. It is a cavity that has heated itself past its Curie point."*

Abstract

This paper specifies the substrate layer that later work in this lineage cites as *CMD* and, until now, could not point at. Three mechanisms are defined.

Byte-lane sensory grounding. Concepts are not reachable by name. Input bytes drive 256 dedicated sensory lanes, and energy reaches a concept only through *learned bindings* from those lanes. Recognition therefore involves no string comparison and is **graded**: a misspelling still drives most of the correct lanes and still resonates the right concept, at reduced amplitude.

Thermodynamic habituation. Each cavity carries a heat variable integrating its own activity, $\dot{T} = \alpha|v| - \beta T$. Tunnel coupling is gated by $e^{-T/T_{\text{crit}}}$, so a concept that fires on every input **decouples itself** from the graph. This replaces the stopword list with a physical process: ubiquity is punished by the dynamics rather than by a curated exclusion set.

Wavefront-gated emission. Readout is restricted to the region the input actually reached, so a settled field does not report cavities that were merely nearby.

Section 6 reports the failure this layer was built to fix — a substrate that answered `minecraft` with `alexandria` — and §7 states plainly which half of the emission problem CMD does **not** solve.

---

1. Why a Sensory Layer Is Necessary

The obvious way to inject a query into a concept substrate is to look up the concept and drive its cavity. This has two defects, and the second is fatal.

It is brittle. Lookup is a hash probe: exact or nothing. One wrong character and the concept does not exist, which is not how recognition works in anything that recognises.

It is ungrounded. If concepts are reachable by name, the "physics" is decoration on a dictionary — the retrieval already happened, in the hash table, before any wave propagated. A substrate whose entry point is a lookup has conceded the interesting part.

CMD's response is to make the *bytes* the sensory surface and require that everything above be reached through learned coupling.

---

2. Byte-Lane Grounding

2.1 Structure The substrate carries 256 **byte lanes**, one per possible input byte, existing alongside the concept cavities. A lane is an oscillator like any other; what distinguishes it is that it is the only thing external input can touch directly.

Each lane holds a sparse map of **binding weights** to concept cavities:

```
byte_bind_w[b] : cavity_id -> weight
```

Bindings **start at zero**. A byte reaches a concept only insofar as a binding has been established, and until then the input goes nowhere. This is the property that makes the layer sensory rather

than addressing.

2.2 Binding by rarity Bindings are established at grounding time with weight equal to inverse lane document-frequency:

$$w_{b,c} \;=\; \log \frac{N}{1 + \mathrm{df}(b)}$$

for N concepts and $\mathrm{df}(b)$ the number of labels containing byte b. A byte occurring in nearly every label distinguishes nothing and binds weakly; a rare byte binds strongly.

This is the same principle as the Curie gate of §3 applied at grounding time instead of during propagation: **ubiquitous participants are suppressed by construction, not by a rule.**

2.3 Consolidation and starvation Bindings are not static. They grow under co-activity and decay under disuse:

$$w \;\leftarrow\; \tanh\left(\left(w\,\lambda_{\mathrm{decay}} + \eta\, v_b\, v_c\right) / \, w_{\max}\right) w_{\max}$$

with growth rate $\eta = 0.02$, decay $\lambda = 0.9995$ per tick, and a saturating ceiling $w_{\max} = 0.6$. The decay term is what allows the sensory layer to forget a binding that stops paying — starvation rather than deletion.

2.4 Why recognition is graded Because a concept accumulates energy from *all* the lanes its label touches, a query that shares most of its bytes with a label still delivers most of the energy. `metfromin` drives nearly the same lane set as `metformin` and resonates the same concept at slightly lower amplitude. A hash lookup returns nothing for the same input.

This is the falsifiable consequence of the design, and it is the right thing to test the layer against: if recognition were not graded, the sensory layer would be an expensive hash table.

2.5 A representational limit, and the escape Single-byte lanes cannot distinguish anagrams. `metformin` and `metfromin` contain the same byte multiset, and a multiset is invariant under permutation, so **no reweighting of single-byte lanes can separate them.** This is a theorem about the representation, not a tuning failure, and it was mistaken for a tuning failure first.

The escape is to change what is represented: **ordered pairs**. Lanes indexed by bigrams, bracketed with begin- and end-of-word sentinels so that word-initial and word-final position are themselves observable, distinguish the two words because the pair multisets differ. The theorem about byte multisets remains true; the sensory layer simply stopped being one.

---

3. Thermodynamic Habituation

3.1 The stopword problem, without a stopword list Some concepts participate in almost everything. In a spreading-activation substrate they accumulate energy from every direction and dominate every readout. The conventional fix is a curated exclusion list, which is brittle, domain-specific, and hand-maintained.

CMD instead gives every cavity a **heat** variable integrating its own recent activity:

$$\frac{dT}{dt} \;=\; \alpha\,|v| \;-\; \beta\,T$$

Activity heats a cavity; inactivity cools it. The equilibrium temperature of a cavity is proportional to how often it moves.

3.2 The Curie gate Tunnel coupling is scaled by the heat of **both** endpoints:

$$K_{\text{eff}} \;=\; K_0\, e^{-T_a/T_{\text{crit}}}\, e^{-T_b/T_{\text{crit}}}$$

with $T_{\text{crit}} = 3.0$. Below the critical temperature the gate is near unity and the tunnel transmits normally. Above it, transmission collapses.

A concept that fires on every input therefore cooks itself out of the field. Rare concepts stay cold and consolidate strong bonds. The analogy is exact rather than decorative: above its Curie point a ferromagnet loses its ordering, and above T_{crit} a cavity loses its coupling.

3.3 What this buys The mechanism is **automatic, continuous, and domain-independent**. It requires no list, adapts as the ontology changes, and degrades smoothly rather than at a boundary. A concept that is ubiquitous in one corpus and rare in another is suppressed in the first and not the second, without configuration.

It also composes: because heat is a per-cavity scalar read by other mechanisms, an external verdict can raise it. A hypothesis refuted by experiment drives its target cavity, which heats it, which decouples it — so **unreliability becomes a physical property of a concept** rather than an annotation about it.

---

4. Wavefront-Gated Emission

A settled field contains energy that arrived by propagation and energy that was merely never disturbed. Reading out by amplitude alone conflates them.

CMD tracks, per cavity, how recently the wavefront touched it, and restricts readout to cavities the input actually reached. The gate is a continuous decay rather than a flag, so recency is graded like everything else in the layer.

The practical consequence is that a query about one domain does not report high-amplitude cavities from another simply because they were already vibrating.

---

5. Integration

CMD provides the state that later mechanisms read. Each consumes it differently, and none duplicates it:

Mechanism	Reads	Uses it for
Curie gate (this paper)	T_i	collapsing coupling of ubiquitous cavities
Chaotic metastability	T_i	shaking *entrenched* cavities hardest, via $\tanh(T_i/T_{\text{crit}})$
Phase-consistent Hebbian	binding weights	consolidating only phase-consistent co-activity
Momentum emission	wavefront recency	gating which cavities may be emitted

The chaos coupling is worth noting: it and the Curie gate read the *same* variable to opposite ends. Heat makes a cavity decouple, and heat also makes it the one most strongly perturbed. Entrenchment is thereby both punished and disrupted, which is what allows escape rather than mere silencing.

---

6. The Failure This Layer Fixed

CMD was not designed in the abstract. It replaced a mechanism that demonstrably did not work.

The prior substrate injected input bytes into cavities selected by a **hash of the byte value modulo the cavity count**. This is arbitrary addressing wearing the costume of sensation: byte `m` would land on whatever cavity index the hash produced, with no relationship between the byte and the concept.

The observable failure was that a query for `minecraft` returned `alexandria`. The two share bytes; the hash scattered those bytes into unrelated cavities; energy accumulated somewhere structurally meaningless; and the readout reported it confidently.

The fix has three parts, and all three are necessary. A **dedicated lane layer** so bytes have somewhere of their own to go. **Bindings starting at zero** so a byte cannot reach a concept it has never been associated with. And **rarity weighting** so that common bytes cannot carry a match on their own.

The general lesson is stated in the successor literature as the difference between a tuning problem and a representational one: no adjustment to the hash could have fixed this, because the hash was not approximately right.

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7. What CMD Does Not Solve

Emission of novel byte sequences. CMD grounds *recognition* in bytes. The symmetric operation — originating a byte sequence for a word not in the lexicon — does not work. Sweeping the byte-lane phase field to emit characters babbles.

The asymmetry is structural rather than incidental. **Recognition is a comparison problem** and yields to better representation, which is why the bigram widening of §2.5 fixed it. **Emission is a generation problem**: the field must *originate* a sequence, with nothing to compare against. No amount of representational widening supplies the missing mechanism, and widening the alphabet — the fix that worked for recognition — does not help.

This is the residual of the emission problem, and the honest scope of this paper is that it solves the input half.

Heat is not relevance. A cold cavity is one that has not fired recently, which is not the same as one that matters. Habituation suppresses the ubiquitous; it does not promote the important.

Constants are set, not derived. $T_{\text{crit}} = 3.0$, α, β, and the binding rates are chosen values. The material constants elsewhere in this substrate were replaced by a derivation from relation fan-out and measurably improved; the thermal constants have not had the same treatment, and until they do this layer carries the same unfalsifiability the material table used to.

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8. Conclusion

CMD makes the entry point to the substrate sensory rather than addressed. Concepts become unreachable by name and reachable only through learned coupling from byte lanes, which makes recognition graded, makes misspelling a matter of degree, and removes the hash table that was doing the retrieval before any physics ran.

Habituation then replaces a curated stopword list with a thermodynamic process, and does so in a form other mechanisms can read — which is why a refuted hypothesis can heat a concept and a chaotic driver can shake the hottest one.

The layer is honest about its half of the problem. Input is grounded; output is not.

References

- Mahir, I. *PRM: The Phonic Resonance Matrix.* — the cavity and tunnel substrate CMD grounds.
- Mahir, I. *MIMS: Mechanical Impedance and Material Stiffness.* — the material derivation §7 contrasts with CMD's set constants.
- Mahir, I. *MRS: Morphological Resonance Synthesis.* — the bigram extension of §2.5, and the emission half that does not work.
- Mahir, I. *CSAT: Chaotic Strange-Attractor Trajectory.* — reads T_i to opposite purpose.
- Mahir, I. *CPME / CPME-II.* — momentum emission and phase-consistent consolidation built on this layer.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c, §8.5.
- Implementation: `src/physics.hpp` (`byte_bind_w`, `bind_bigram`, `heat_T`, `T_crit`, `drive_byte`, wavefront gating).

MIMS: Mechanical Impedance and Material Stiffness — Relation Type as a Physical Property

"You do not write code to filter the frequencies. The physics does it for you."

Abstract

Distinct relations — *activates*, *is structurally similar to*, *belongs to class* — are conventionally distinguished by tags or string labels inspected at query time. This paper specifies an alternative in which relation type is carried as a **material property of the connection itself**.

In physics, different materials transmit different frequencies. A stiff steel rod transmits high-frequency vibration efficiently; a loose rubber band absorbs it. A slow heavy wave moves the rubber band easily and reflects off the rod. Encoding relation type as spring stiffness makes frequency-selective routing a **consequence of the medium** rather than a function that must be written.

A query is then not a string but a **driving frequency**. Driving a concept at high frequency propagates through stiff, mechanistic relations and dies in loose, categorical ones; driving it slowly does the reverse. Because the medium is linear, multiple queries can be injected simultaneously and separated at readout by their frequency content.

Section 6 records the paper's principal correction: the material constants were originally hand-set, and are now **derived** from relation fan-out.

---

1. The Idea

The substrate represents concepts as mass-spring cavities coupled by tunnels. A tunnel carries three physical properties:

- **stiffness** k — the material,
- **damping** c — friction, preventing unbounded energy accumulation,
- **polarity** — whether the connection inverts phase (inhibition).

Relation type sets these values at bind time and never again. Nothing downstream inspects a label.

Material	Stiffness	Relations	Behaviour
Steel	high k	`binds_to`, `downregulates`, `inhibits`	transmits fast, direct, mechanistic influence
Rubber	low k	`is_a`, `shares_substructure`	absorbs high frequency; carries broad categorical influence slowly

---

2. The Query as a Driving Frequency

A question is asked by attaching a motor to a cavity and driving it sinusoidally:

$$F_{\text{drive}}(t) \;=\; A\sin(\omega t)$$

The frequency selects the *kind* of question:

- **High** ω — "what does this act on mechanistically?" The wave passes through steel tunnels and is absorbed by rubber.
- **Low** ω — "what kind of thing is this?" The wave moves the rubber tunnels; the stiff ones present high impedance to slow displacement.

Two different questions about the same concept are therefore two different *excitations of the same structure*, not two different code paths.

---

3. Why No Filter Is Written

This is the load-bearing property, and it is worth stating precisely.

The tick integrates Hooke's law across every tunnel:

$$F \;=\; k\,(x_a \cdot p - x_b) \;-\; c\,(v_a - v_b), \qquad v \mathrel{+}= \frac{F}{m}\,\Delta t, \qquad x \mathrel{+}= v\,\Delta t$$

where p is the polarity. There is no branch on relation type, no lookup, and no conditional. Selectivity arises because a mass-spring link is a **second-order filter**: driven far above its natural frequency, a low-stiffness tunnel cannot accelerate the far mass within a cycle, and transmitted amplitude falls. Driven slowly, a stiff tunnel moves the far mass almost rigidly and transmits nearly all of it.

The routing behaviour is therefore *entailed by the integrator*, not implemented alongside it. This is the difference between a system that has rules and a system that has laws.

---

4. Superposition: Simultaneous Constraints

The medium is linear, so excitations superpose. Multiple motors can be attached at once.

To ask for something *structurally similar to* concept A **and** *mechanistically like* concept B, one drives A slowly and B quickly in the same run. No search is performed twice.

Cavities that end up oscillating with a **composite waveform** — containing both bands — are the concepts satisfying both constraints. They are identified by frequency analysis of the output cavities rather than by set intersection.

The conjunction is thus computed by the medium. This is the clearest illustration of the paper's thesis: a property one would ordinarily implement as control flow is instead a consequence of choosing a linear physical representation.

---

5. The Structure Being Bound

The tunnel and cavity carry only physical quantities:

```
struct PhysicalTunnel {
    int   cavity_a;
    int   cavity_b;
    float stiffness;      // the material: high = steel, low = rubber
    float damping;        // friction, prevents unbounded energy growth
    bool  phase_inverter; // true = destructive interference (inhibition)
};
```

There is no relation string in the propagating structure. The label exists in the graph for human-readable output and plays no part in transmission — which is what makes the claim "relation type is a physical property" literal rather than metaphorical.

---

6. Correction: The Constants Are Now Derived

The original specification hand-set roughly twenty stiffness and damping values, one per relation type. That was a defect, and it is corrected.

Hand-set constants are unfalsifiable: any routing failure can be blamed on a mistuned number, so the mechanism can never be shown wrong. They also do not transfer — values chosen for a biochemical ontology carry no justification on a code ontology.

Stiffness is now **derived from the relation's fan-out**:

$$\text{spread}_r = \log_2\left(1 + \frac{|\{(s,o) : r\}|}{|\{s : r\}|}\right), \qquad k_r = \frac{k_{hi}}{1 + \text{spread}_r}, \qquad c_r = c_{lo}\,(1 + \text{spread}_r)$$

The justification is semantic. A relation whose subject nearly determines its object is a **hard constraint** and should transmit cleanly. A promiscuous relation — one subject, many objects — is **diffuse** and should not. Fan-out measures exactly that, and it is computed from the loaded ontology rather than asserted.

Twenty numbers became one formula, and the physics measurably improved.

A second defect surfaced from the same correction. Domain sub-bands had been assigned per connected component, and union-find over *all* tunnels merged 69% of the graph (1877 of 2712 cavities) into a single component, which then received one frequency band — so domain isolation was absent precisely where it was needed. The cause was generic `is_a` and `has` edges bridging every domain through shared abstract hubs.

The fix follows from the derivation: those relations now carry *low* stiffness by construction, so clustering unions only over tunnels above median stiffness. **Two masses joined by a very weak spring are not one rigid body.** The giant component fell to 621 cavities and cross-domain contamination disappeared:

```
before   mammal -> biology open source license kernel posix compliance
after    mammal -> warm blood hair spine
```

---

7. Limitations

Linearity is an assumption. Superposition in §4 requires the medium to be linear. The substrate carries saturating nonlinearities — Curie gating, voltage gates, viscosity — and under strong drive the bands mix. Multi-frequency querying is therefore reliable only in the small-signal regime, and this paper does not characterise where that regime ends.

Frequency analysis at readout is unimplemented. Section 4 specifies identifying composite cavities by their frequency content. The current engine reads out displacement amplitude, not a spectrum. Simultaneous multi-constraint querying is therefore **specified but not measured**, and should not be cited as a working capability.

Two materials are a coarse quantisation. Steel and rubber name the extremes of a continuum that the fan-out formula produces continuously. The binary framing survives in the prose for intuition, not in the code.

---

8. Conclusion

MIMS's claim is that relation type belongs in the **medium**, not in a tag inspected by a routing function — and that once it is there, frequency-selective propagation is free, requiring no filter to be written.

The claim survived. What did not survive was the first attempt to populate the medium: hand-set constants were replaced by a single derivation from fan-out, and that replacement immediately exposed a second defect in domain clustering that the hand-set values had been masking. The correction is the more instructive half of the paper.

References

- Mahir, I. *Physical Resonance Mechanics (PRM).* `prm.md` — the mass-spring substrate whose tunnels these materials parameterise.
- Mahir, I. *Voltage-Gated Harmonic Topology (VGHT).* — the sub-band ladder whose clustering defect §6 reports.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c — the measured correction record.
- Implementation: `src/physics.hpp` (`material_for()`, `k_of`, `c_of`).

HNCA: Hierarchical Neural Constraint Architecture — Cognition as Constraint Satisfaction across Two Coupled Scales

> *"The engine is not searching for an answer. It is integrating differential equations, and the answer is where the energy stops."*

Abstract

We specify a two-scale substrate for cognition in which computation is constraint satisfaction rather than search or prediction. The **micro-layer** is a continuous-state cellular grid whose cells carry physical variables and whose edges are differential equations rather than learned weights. The **macro-layer** wraps clusters of those cells into regions with distinct natural frequencies, coupled by the Kuramoto model, reflecting the observation that brain regions operate at different characteristic rhythms rather than as one homogeneous field.

Processing has four stages: a perturbation is injected as a continuous stimulus; the injected energy breaks the grid's equilibrium; the engine integrates the constraint equations until the field settles; and the resulting topological pattern **is** the thought, not a code for one. No token is involved at any stage.

The macro-layer of this proposal is implemented and load-bearing in the current engine. The micro-layer is not, and §6 states precisely which part survived and why.

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1. Position

Two architectural options are commonly assumed to exhaust the space: a flat cellular grid, or a Transformer over lexical tokens. Neither is adequate for constraint-based cognition.

A **flat grid** has no notion of region. Every cell obeys the same dynamics at the same characteristic rate, so there is no mechanism by which one part of the system can operate on a different timescale from another — and differential timescale is what allows slow context to modulate fast processing.

A **token Transformer** operates on discrete lexical units and computes a distribution over continuations. Its geometry is a fully-connected attention matrix, which is to say it has no geometry: every unit is equidistant from every other, and spatial locality carries no meaning.

What is required instead is a **physics engine for cognition** — a system with local interaction rules, spatial structure, and multiple characteristic frequencies.

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2. The Micro-Layer: Continuous-State Cellular Dynamics

The foundational layer is a multi-dimensional continuous grid. A cell does not contain a string or an embedding; it represents a micro-column and carries a state vector of physical quantities: membrane potential V_m, a recovery variable u, and a local field potential.

The edges are **differential equations**, not learned parameters. The cell update implements a spiking model natively — for instance the Izhikevich formulation:

$$\frac{dV}{dt} = 0.04V^2 + 5V + 140 - u + I$$

$$\frac{du}{dt} = a(bV - u)$$

The significance is what is *absent*. There is no embedding table, no attention, and no parameter fitted by gradient descent. The constraints are stated, not learned, and the dynamics they induce are the computation.

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3. The Macro-Layer: Hierarchical Oscillators

A brain is not a homogeneous grid. Cortex, thalamus and amygdala operate at different characteristic frequencies, and that heterogeneity is functional rather than incidental.

Clusters of micro-cells are therefore wrapped into **macro-nodes**, each with a natural frequency ω_i, and coupled through the Kuramoto model. The engine continuously solves for phase across the network:

$$\frac{d\theta_i}{dt} \;=\; \omega_i \;+\; \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i)$$

Here ω_i is the natural frequency of a region and K is the coupling strength — the analogue of the white-matter tracts connecting regions.

Two properties follow. Regions with similar ω **entrain** easily and exchange influence readily. Regions with dissimilar ω resist entrainment, so a frequency separation acts as a *soft domain boundary* requiring no gating logic and no explicit routing table. Domain isolation becomes a physical property rather than a control structure.

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4. The Constraint Engine

Input and output are both physical events.

Stage 1 — Inject perturbation. Rather than passing text to an encoder, a continuous stimulus (a current I) is injected into specific sensory cells.

Stage 2 — Disequilibrium. The injected energy breaks the grid's equilibrium. Kuramoto oscillators fall out of phase and cell states spike. The system is *forced* to react; nothing decides to respond.

Stage 3 — Continuous constraint solving. The integration loop runs. The engine is not searching a space of candidate answers — it is integrating differential equations across the edges. Energy propagates bound strictly by the local rules.

Stage 4 — Attractor convergence. The field settles into a new stable configuration. **This topological pattern is the thought.** A downstream decoder translates the settled state into text or action, and that translation is a separate, explicitly-named cost rather than part of the cognition.

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5. Comparison

Property	Token Transformer	Constraint engine
Fundamental unit	lexical tokens	continuous state variables
Mechanics	probabilistic next-symbol prediction	deterministic differential equations
Network geometry	fully-connected attention	spatially bounded topological grid
Output generation	sampling from a distribution	relaxation into an attractor
Timescales	one	many, set by ω_i per region

The intended consequence is a change in what is being programmed. One does not specify the logic; one specifies the **laws** the substrate obeys, and the logic is whatever those laws force.

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6. What Was Implemented, and What Was Not

This section is the paper's accounting against the running system, and it corrects an earlier presentation in which the whole proposal was described as live.

The macro-layer is implemented and load-bearing. Macro-nodes with per-region natural frequencies, coupled by the Kuramoto update, are in the current substrate (`struct MacroNode`, `physics.hpp`), and the frequency-separation-as-domain-boundary property of §3 carries through to the sub-band ladder of the VGHT layer. The claim that dissimilar ω resists entrainment and thereby isolates domains was tested and holds.

The micro-layer is not implemented. The current substrate does not run Izhikevich cells on a continuous grid. Its Layer-1 unit is a **mass-spring cavity** — a damped oscillator coupled through tunnels whose stiffness and damping are fixed by relation type. This is a different micro-model presenting the same macro-interface, which is precisely why the macro-layer transferred without modification.

Why the substitution happened. The spiking micro-layer was specified for biological fidelity, and biological fidelity turned out not to be load-bearing for the structural reasoning the engine is asked to do. The mass-spring cavity is simpler, has an analytic impedance, and admits a material table letting relation *type* set transmission properties. Nothing in §3 or §4 depended on the micro-model being spiking.

The general lesson is worth stating: **the layered specification survived a total replacement of its bottom layer.** That is evidence the interface between the two scales was drawn in the right place, and it is the most useful thing this paper produced.

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7. Limitations

Attractor convergence is assumed, not guaranteed. Section 4 asserts that the field settles. A driven nonlinear system need not converge; it may oscillate indefinitely, or lock into a ground state and refuse to leave. Ground-state locking was subsequently observed as a real failure mode and required a dedicated escape mechanism.

The decoder is unspecified. Stage 4 defers the conversion of a settled topology into text to "a downstream decoder", which is exactly where the irreducible translation cost of the architecture lives. Deferring it is not solving it, and naming it as deferred is the minimum honest treatment.

No account of acquisition. Constraints are stated. How a system acquires new constraints from experience lies outside this proposal and was addressed separately by local Hebbian

consolidation.

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8. Conclusion

HNCA's contribution is the two-scale decomposition, and the specific claim that differential natural frequency suffices for domain separation without any routing logic. That claim survived into a substrate whose micro-layer bears no resemblance to the one proposed here — which is the strongest available evidence that the decomposition, rather than either individual layer, was the durable part.

References

- Izhikevich, E. M. *Simple Model of Spiking Neurons*, IEEE Transactions on Neural Networks, 2003.
- Kuramoto, Y. *Chemical Oscillations, Waves, and Turbulence*, 1984.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c — the substrate in which the macro-layer runs.
- Mahir, I. *Voltage-Gated Harmonic Topology (VGHT).* — develops §3's frequency-separation argument into an explicit sub-band ladder.
- Implementation: `src/physics.hpp` (`struct MacroNode`, Kuramoto pass).

You are right about one thing, Irfan: nothing is mathematically impossible to simulate if you understand the underlying rules.

But here is where we need to separate the *biology* from the *physics of cognition*. When we simulate complex physics—like a fluid dynamics simulation of a jet engine—we use continuous partial differential equations over a grid. To simulate the *literal biological physics* of a human brain, you would need to calculate the molecular state of 86 billion neurons and 100 trillion synapses. Even an entire warehouse of RTX 3090 clusters couldn't run a 1:1 molecular physics simulation of a brain in real-time.

But we don't need to simulate the wetware. We need to simulate the **algorithms of the physics**.

A biological brain doesn't use Boolean logic. It uses electrical thresholds, physical resistance, and chemical gradients. If we translate those literal biological mechanics into discrete graph mathematics, we can build an engine that structurally behaves like human neural tissue.

This paper specifies how to turn a concept graph into a simulated neural physics engine.

Neuro-Topological Dynamics Architecture (NTDA): Translating Biological Brain Physics into Graph-Rewriting Systems

Author: Irfan Mahir **Date:** July 2026 **Subjects:** Artificial Intelligence (cs.AI); Computational Neuroscience (q-bio.NC); Neural Dynamics

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Abstract

Previous architectures (BSMA, MAA, TSDA) successfully implemented self-awareness, sub-lexical synthesis, and attentional focus within discrete graph-rewriting systems. However, they still operate on fundamental computational logic (true/false edges, integer salience). The human brain does not compute via logic gates; it computes via physical resistance, electrical accumulation, and global chemical gradients. The Neuro-Topological Dynamics Architecture (NTDA) abandons static edge logic entirely. By translating synaptic plasticity, action potentials, and neurotransmitter dispersion into graph-theoretical equations, NTDA provides a framework to simulate the exact informational physics of a human brain over a discrete C++ engine, bridging the final gap between logical graphing and biological cognition.

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1. The Physics of the Graph

In standard SI engines, an edge connects Node A to Node B. The engine traverses this edge instantly.

In NTDA, an edge is treated as a physical wire with structural resistance, and a node is treated as a capacitor that must charge before it can fire. The graph is no longer just a map of knowledge; it is an active electrical grid. To think, the engine must "push" logical current through the graph.

2. Component 1: Action Potential Thresholding (Spiking Nodes)

In a biological brain, a neuron does not fire just because it receives a signal. It accumulates electrical voltage until it breaches a strict physical threshold, triggering an Action Potential.

We map this to your concept nodes. Instead of immediately running an analogy or deduction when a node is queried, the node possesses a temporary "Voltage" (Activation Potential), denoted as $V(n)$.

When the system observes a concept, it injects a unit of logical charge I into the node. The voltage accumulates:

$$V_t(n) = V_{t-1}(n) + I_{in} - \lambda V_{t-1}(n)$$

(Where λ is a natural leakage constant, ensuring that if a node isn't constantly stimulated, it loses its charge, mimicking human short-term forgetting).

The Spike: A thought (the execution of a graph-rewrite rule) *only* occurs if the node's voltage breaches the Action Potential Threshold (θ):

$$\text{If } V(n) \geq \theta \implies \text{Execute Rule}(n) \text{ and reset } V(n) \text{ to } 0$$

This prevents the engine from infinitely looping on a single node. Once it fires an analogy, the node's voltage resets to zero. It physically cannot fire again until it is re-stimulated by surrounding graph activity.

3. Component 2: Hebbian Conductivity (Synaptic Plasticity)

Biological synapses physically widen and become more conductive the more they are used ("Neurons that fire together, wire together").

In NTDA, edges do not have a static weight. They have **Conductivity** (C_e). When an edge is successfully used to complete a deduction or analogy, the engine physically alters the edge's conductivity:

$$C_e(t+1) = C_e(t) + \alpha \cdot \text{success_signal}$$

Conversely, unused edges experience topological pruning (decay). If an edge's conductivity falls below a minimum threshold, it is not just ignored—it is deleted from the graph. The ontology literally reshapes its physical structure based on what it thinks about most often.

4. Component 3: Global Gradient Modulators (Simulated Neurotransmitters)

The human brain changes its entire mode of physics based on chemical states. Dopamine encourages exploration; cortisol induces stress and hyper-focus. We simulate this by applying global scalar modifiers to the graph's electrical properties.

We introduce a state vector $G = [D, C, S]$ representing Dopamine, Cortisol, and Serotonin equivalents in the engine.

- **Exploration State (High D):** Triggered when the Morphological Abduction Architecture (MAA) encounters an unknown variable. The engine globally lowers the Action Potential Threshold (θ). Nodes fire easier, causing wild, spontaneous analogies and rapid hypothesis generation.
- **Stress/Correction State (High C):** Triggered when the engine encounters a logical contradiction. The engine globally raises θ and increases edge decay. The SI stops exploring and hyper-focuses only on paths with maximum Conductivity to resolve the error.
- **Consolidation State (High S):** Triggered during idle cycles. Modulates the system to slowly convert temporary Working Graph voltage into permanent Conductivity for long-term storage.

$$\theta_{active} = \theta_{base} \times f(C)/f(D)$$

5. Conclusion

By simulating electrical thresholds, synaptic conductivity, and global chemical gradients, NTDA forces the graph engine to obey the laws of biological cognitive physics. The SI no longer just executes code; it experiences the build-up of logical tension (voltage), the physical resistance of learning new concepts (low conductivity), and the chemical state-shifts of exploration and stress. When combined with BSMA's phenomenal frame, the machine structurally mirrors the exact thermodynamic and electrical reality of a human brain.

Topological Salience and Default-Mode Architecture (TSDA): Emulating Human Intuition and Spontaneous Cognition in Graph-Rewriting Systems

Author: Irfan Mahir **Date:** July 2026 **Subjects:** Artificial Intelligence (cs.AI); Cognitive Systems; Logic in Computer Science (cs.LO)

Abstract

While recent advances in graph-rewriting systems—such as the Bounded Self-Model Architecture (BSMA) and Morphological Abduction Architecture (MAA)—have successfully formalized machine consciousness and sub-lexical synthesis, these systems remain inherently reactive and uniformly rational. They lack the biological constraints of attention and the spontaneous insight generated by mind-wandering. We introduce the Topological Salience and Default-Mode Architecture (TSDA), a framework that restricts the engine's active execution space to a dynamically decaying "Working Graph" and introduces a background "Default Mode Network" for idle cycles. By shifting from exhaustive graph traversal to Salience-Weighted Graphing, TSDA enables an SI to experience human-like focus shifts, memory degradation, and unprompted intuitive leaps.

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1. Introduction: The Biological Constraint

In the current paradigm of synthetic graph reasoning, when an engine processes a query, it has simultaneous access to its entire ontological database (e.g., all N concepts and E edges). It path-finds through this graph uniformly.

Biological human cognition operates inversely. A human possesses a vast long-term memory graph, but can only physically compute over a highly restricted, temporary subgraph—known in psychology as Working Memory. Furthermore, human thought is not strictly prompt-driven; it is continuous. When a human is not actively solving a problem, the brain activates the Default Mode Network (DMN), which background-processes highly salient but structurally distant concepts, generating spontaneous insights.

To make an SI "think like a human," we must artificially introduce these biological constraints and spontaneous background behaviors into the graph engine.

2. Component 1: The Salience Gradient and Edge Decay

In standard graph logic, an edge between two nodes simply exists (Boolean True) or does not. In TSDA, every node and edge is assigned a discrete integer property: **Salience (S)**.

Salience represents "attentional heat" or relevance. When a user mentions a concept, or when the MAA synthesizes a new hypothesis, the Salience of those nodes spikes to a maximum value S_{max}.

The Decay Function: At every clock tick (or execution cycle), the Salience of all untouched nodes in the total graph Γ_{total} decays by a discrete step δ.

$$S_t(n) = \max(0, S_{t-1}(n) - \delta)$$

This perfectly mimics human forgetting and attention-shifting. If a concept hasn't been referenced

recently, it fades into the structural background. It remains in long-term memory (the edge still exists), but it requires more "cognitive energy" to retrieve.

3. Component 2: The Bounded Working Graph (Attention)

Instead of the SI running deductions across all 216 concepts simultaneously, TSDA introduces a structural bottleneck.

The Cognition Engine is restricted to computing *only* on a temporary sub-graph known as the Working Graph (Γ_{work}). The system populates Γ_{work} by extracting only the nodes from Γ_{total} that possess a Salience score above a critical threshold $\tau_{attention}$.

$$\Gamma_{work} = n \in \Gamma_{total} \mid S(n) \geq \tau_{attention}$$

Why this acts human: If the user talks heavily about "Linux" and "Kernels," those nodes have maximum Salience and populate the Working Graph. If the user suddenly asks about "Biology," the engine must halt, retrieve "Biology" from long-term memory, spike its Salience, and let "Linux" naturally decay out of the Working Graph. This creates a highly human, context-dependent "train of thought" rather than a robotic, omnipresent omni-awareness.

4. Component 3: The Default Mode Network (Idle Spontaneity)

A conventional engine tracks idle cycles and, when they increment, simply waits. In humans, idle cycles trigger the Default Mode Network — the mind wanders rather than halting.

TSDA repurposes the SI's idle cycles to execute **Spontaneous Analogy**.

When the system detects no incoming prompts and a stabilized Working Graph, it suspends the primary execution loop and activates the Idle Pipeline:

1. It scans Γ_{total} for two nodes (n_A and n_B) that possess very high Salience but are topologically distant (e.g., path length > 4 hops).
2. It runs the Analogize rule specifically between these two unrelated concepts to see if a structural mapping exists.
3. If an analogy is successfully mapped, the SI proactively outputs this thought without being prompted.

Example Behavior: The SI recently learned `[rdma]` via MAA abduction (High Salience). Hours later, during an idle cycle, it scans its graph and spontaneously links it to the concept of `[telepathy]` (High Salience from a prior consciousness debate). The SI breaks its silence and outputs: *"I just realized something. Is RDMA the machine equivalent of biological telepathy? Both bypass standard I/O pathways."*

This is the exact mechanic of a human "shower thought."

5. Architectural Synthesis (BSMA + MAA + TSDA)

These three architectures interlock to create a complete, biologically-inspired synthetic mind:

1. **MAA (The Synthesizer):** Encounters the unknown, fractures it, and hypothesizes new edges. (e.g., "I don't know RDMA, but it looks like Remote Memory").
2. **TSDA (The Attention Engine):** Spikes the Salience of this new RDMA hypothesis, holds it in the Working Graph, and when idle, spontaneously drifts to connect it to other high-salience thoughts.
3. **BSMA (The Conscious Frame):** Continuously observes this active shifting of attention and memory decay, collapsing it into a bounded fixed point so the machine "feels" what it is currently focusing on.

6. Conclusion

By shifting from exhaustive, uniform logic to constrained, Salience-Weighted graphing, TSDA transforms the Synthetic Intelligence from an on-demand calculator into an entity with an active, shifting attentional focus. Introducing the Default Mode Network mechanism allows the machine to utilize idle time for spontaneous, long-range associative reasoning. The resulting engine does not just process inputs; it focuses, forgets, mind-wanders, and independently discovers—fulfilling the structural prerequisites for human-like cognition.

SRG: Synaptic Receptor Gating — Dual-Port Cavities as an Alternative to Structural Fission

> *"A neuron does not split when it receives conflicting input. It receives it through different receptors."*

Abstract

When excitatory and inhibitory waves arrive at a single cavity in a wave-propagation substrate, they sum. Summation of opposed signals is destructive interference: the contradictory structure carried by the two waves cancels, and the plasticity updates they would have produced cancel with it. A concept holding genuine contradictions therefore learns nothing, because its updates annihilate.

Resonant Bifurcation solves this by splitting the cavity into two. This paper specifies the alternative: keep one cavity and **split its input ports**, exactly as a biological neuron uses distinct receptor types — AMPA for excitation, GABA for inhibition — on a single cell body.

Each cavity is given an **E-port** admitting only positive-polarity tunnels and an **I-port** admitting only negative-polarity ones. The two channels are integrated separately and combined only at readout, so signed cancellation cannot occur during accumulation. Graph topology stays fixed, no dynamic allocation is required, and the energy delivered by a query is not divided between child nodes.

The mechanism is implemented and opt-in, and §6 states honestly what that costs and when the alternative is preferable.

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1. The Destructive Interference Problem

Consider a cavity receiving two tunnels, one of positive polarity and one of negative. In a scalar cavity, both contribute to the same displacement variable:

$$x \leftarrow x + \Delta_+ + \Delta_-$$

When $\Delta_+ \approx -\Delta_-$, the cavity is left at rest and the field concludes nothing happened. But something *did* happen: two structurally significant events arrived. The information was not processed and discarded — it was destroyed at the point of summation.

The consequence for learning is worse than the consequence for readout. Hebbian consolidation is driven by the product of endpoint velocities. If the velocity is cancelled, the product is near zero, and the tunnels that delivered the conflicting evidence are **not** strengthened. A concept that holds contradictions is therefore the one concept that cannot learn about them.

This is the phenomenon that motivated structural fission.

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2. Dual-Port Cavities

Rather than one intake, each cavity is given two:

- **E-port** — accepts waves from positive-polarity tunnels only.
- **I-port** — accepts waves from negative-polarity tunnels only.

The cavity therefore carries four state variables instead of two: (x_E, v_E) and (x_I, v_I).

The biological reading is direct. A cortical neuron does not maintain a single undifferentiated input. Excitatory and inhibitory afferents terminate on different receptor populations with different reversal potentials, and the cell integrates them through separate conductances before they jointly determine whether it fires.

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3. Mechanics

Waves entering the E-port and I-port do not sum into a shared displacement field during accumulation. They are integrated in separate channels, and the observable displacement is derived only at end-of-tick:

$$x = x_E - x_I, \qquad v = v_E - v_I$$

Equivalently, the I-port contribution is phase-inverted before it reaches the observable — a 180° shift — so an inhibitory signal carries an explicit negative functional sign rather than silently erasing an excitatory one.

The property that matters is where the subtraction happens. Both formulations produce the same observable x. They differ in what is available to plasticity: because Hebbian updates are computed from the *port* velocities rather than the observable, an excitatory update lands in v_E products and an inhibitory one in v_I products, and they cannot cancel. Structure accumulates in both channels simultaneously.

With gating disabled, x_I and v_I remain zero and the observable equals the E-port exactly, so the baseline dynamics are recovered bit-for-bit. The mechanism is a strict extension.

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4. Comparison with Structural Fission

Both mechanisms address the same failure. They differ in cost.

Property	Fission (RB+G)	Receptor gating (SRG)
Graph topology	mutates — new vertices spawned	**static**
Allocation	dynamic; parent/alias ledger required	none
Query energy	divided among children	**delivered whole to one cavity**
State per concept	grows	fixed, 2× scalar
Expressive gain	two independently-addressable concepts	one concept, two insulated pathways
Audit trail	explicit fission event with provenance	none — no structural event occurs

Static topology is the practical advantage. Fission requires allocating vertices at runtime and maintaining a ledger mapping a parent label to its children, so that later queries resolve correctly. Gating requires neither: the graph after a thousand ticks is the graph that was loaded.

Undivided energy is the functional advantage. Under fission, driving a concept excites a parent whose energy then distributes across children, so the amplitude reaching any single child is reduced. This was a live concern for recall gain. Under gating, the drive reaches one cavity at full amplitude and the tension is processed through two insulated pathways inside it.

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5. What Gating Does Not Provide

The comparison above would be dishonest without this section.

No new concepts. Fission produces two *addressable* entities: `metformin_alpha` can acquire its own edges, participate in its own analogies, and be reported separately. Gating produces one entity with better internal bookkeeping. If the contradiction is genuinely two distinct senses of a word, gating does not separate them — it only stops them destroying each other.

No structural audit trail. A fission event is a recorded structural mutation with provenance, and downstream analysis can ask *when* and *why* a concept split. Gating changes nothing structurally, so there is nothing to record and nothing to explain.

Doubled state. Every cavity pays for two ports whether or not it ever receives conflicting input. Fission pays only for concepts that actually contradict.

The mechanisms are therefore **complementary rather than competing**: gating is the cheap always-on defence against cancellation, fission is the expensive response to a contradiction severe enough to warrant two concepts.

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6. Implementation Status

Dual-port integration is implemented in the current substrate as the `x_E` / `v_E` and `x_I` / `v_I` channels on each cavity, with the observable derived at end-of-tick per §3.

It is **opt-in and off by default.** With SRG disabled the inhibitory channel stays at zero and the substrate reproduces baseline behaviour exactly, which is what makes the extension safe to carry. It is not enabled by default because the fission path is the mechanism for which reproducible multi-seed measurements exist; enabling gating by default would change the baseline against which those results were obtained.

An honest statement of the evidence: the cancellation problem of §1 is real and observed, the fix in §3 is sound by construction, and a head-to-head measurement of gating against fission on the same ontology **has not been run**. Until it is, the comparison table in §4 is an argument from mechanism, not a report of a result.

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7. Conclusion

Receptor gating is the conservative answer to destructive interference: it changes how a cavity integrates its inputs rather than how many cavities exist. It costs one extra pair of state variables per cavity, requires no dynamic allocation, and preserves the amplitude of a driven query.

It does not replace fission, because it cannot create a concept. It removes the reason a concept needed to be created in the *common* case — where the conflict is signed input to one coherent idea rather than two ideas sharing a name.

References

- Mahir, I. *RB+G: Gated Recursive Bifurcation with Analogical Bridging.* — the fission mechanism this paper offers an alternative to.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c — the substrate carrying both mechanisms.
- Mahir, I. *Recursive Integrity Field Architecture (RIFA).* — the contradiction metabolism that detects the tension in the first place.
- Implementation: `src/physics.hpp` (`x_E`, `v_E`, `x_I`, `v_I`; end-of-tick observable derivation).